

SHORT BIO. My research focuses on causal decision making (NeurIPS-25, Preprint, ICLR-26) and causal effect identification (NeurIPS-24). I investigate how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations. In particular, I develop methods for robust decision-making in settings under environmental uncertainty (NeurIPS-25), domain shifts between training and deployment environments (Preprint), and even scenarios involving counterfactual actions (ICLR-26).

EDUCATION

Graduate School of Data Science, Seoul National University Seoul, Republic of Korea
 Integrated M.S. and Ph.D. in Data Science Sep. 2022 – Feb. 2027
 Advisor: Prof. Sanghack Lee.
 Thesis: “Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals”.
 Overall GPA: 4.02 / 4.3; Ph.D. Qualification Exam partially waived due to outstanding academic performance.

Department of Mathematics and Statistics, Sejong University Seoul, Republic of Korea
 B.S. in Applied statistics and Mathematics (Double major), *Cum Laude* Mar. 2016 – Aug. 2022
 Advisors: Prof. Ji Eun Lee and Jin Hee Yoon. (2.5 years military service)
 Overall GPA: 4.04/4.5; Major GPA: 4.46 (Applied Statistics), 4.38 (Mathematics).

PUBLICATIONS

* for joint first authorship and † for corresponding author.

- ◇ Min Woo Park and Sanghack Lee†. [Counterfactual Structural Causal Bandits](#). *International Conference on Learning Representations (ICLR)*, 2026.
- ◇ Min Woo Park, Andy Ardit, Elias Bareinboim†, Sanghack Lee†. [Structural Causal Bandits under Markov Equivalence](#). *Neural Information Processing Systems (NeurIPS)*, 2025, Columbia CausalAI Laboratory, Technical Report (R-122)
- ◇ Min Woo Park and Sanghack Lee†. [On Transportability for Structural Causal Bandits](#). *arXiv preprint*, 2025. under review.
- ◇ Yonghan Jung†, Min Woo Park, and Sanghack Lee†. [Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference](#). *Neural Information Processing Systems (NeurIPS)*, 2024.
- ◇ Min Woo Park*, and Ji Eun Lee*†. [Computation of the iterated Aluthge, Duggal, and Mean transforms](#). *Filomat*, 2023.
- ◇ Min Woo Park*, Taehui Yun*, Youngin Jang, Younsuk Yeom, Jonghwan Kim, LG AI Research, and Sanghack Lee†. Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery. under review.
- ◇ Yesong Choe, Yeahoon Kwon, Min Woo Park, and Sanghack Lee†. Canonical Domain Reduction for Partial Counterfactual Identification. under review.
- ◇ Ji Eun Lee*† and Min Woo Park*. Convergence of the iterated mean transforms of a 2×2 matrix. under review.

AWARDS AND HONORS

Graduate Student Instructor (GSI) scholarship Spring 2024, Spring 2025, Spring 2026
BK21 scholarship Fall 2024, Fall 2025
First place merit scholarship | Sejong University Fall 2020, Fall 2021
Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University
 ▷ Mitigating spurious regression in time-series data via Granger causality. Nov. 2021
Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University
 ▷ The significance of the Euler’s number and its applications. Nov. 2020

RESEARCH EXPERIENCE

Furiosa AI AI Research Engineer Internship	Jun. 2026 – Nov. 2026, scheduled
Causality Lab, Seoul National University Ph.D. Researcher	Jan. 2023 – Feb. 2027
Functional Analysis Lab, Sejong University Undergraduate Researcher	Jul. 2021 – Dec. 2022

PROJECTS

Towards More Scalable Score-based Causal Discovery LG AI Research	Sep. 2025 – Aug. 2026
▷ Leading a project on scalable score-based causal discovery algorithms. (First-author submission under review.)	
- Developed a novel operator enabling score-based causal discovery algorithms to escape local optima.	
- Investigating parallelization strategies for score-based causal discovery algorithms.	
Hyundai NGV Data Boot-Campus 2025 Hyundai Motor	Jul. 2025 – Sep. 2025
▷ Developed high-voltage battery performance prediction technology.	
Hyundai NGV Data Boot-Campus 2024 Hyundai Motor	Jul. 2024 – Sep. 2024
▷ Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometrybased automatic curvature analysis.	
Deep Reinforcement Learning for Recommendation Systems Coursework Project	Mar. 2024 – Jun. 2024
▷ Developed a recommendation system based on DDPG, leveraging multi-modal item features (video, audio, and text) to improve recommendation performance.	
- Source code available at github.com/MinwooPark96/ddpgRLRS .	
Optimal Mixed Policies Completeness Oxford University	Aug. 2023 – Feb. 2024
▷ Studied complete characterization of conditions under which the intervention effect of a variable is positive.	
Acknowledged for helpful comments in Toward a Complete Criterion for Value of Information in Insoluble Decision Problems . <i>Transactions on Machine Learning Research</i> (TMLR), 2024.	

TEACHING ASSISTANTS

Causal Inference for Data Science	Spring 2024, Fall 2024, Spring 2025, Spring 2026
▷ Topics included Bayesian networks and graphical causal inference.	
Machine Learning and Deep Learning for Data Science II	Fall 2023
▷ Covered kernel methods, Gaussian processes, and related topics in machine learning.	
Basic Mathematics for Artificial Intelligence (Undergraduate)	Spring 2022
▷ Covered linear algebra, basic statistics, and probability theory.	

ACADEMIC SERVICES, PROFESSIONAL TALKS AND POSTERS

Counterfactual Structural Causal Bandits	
ICLR 2026 poster session Rio de Janeiro, Brazil	Apr. 2026
Structural Causal Bandits under Markov Equivalence	
ExploreCSR poster session Seoul, Republic of Korea	Dec. 2025
NeurIPS 2025 poster session San Diego, USA	Dec. 2025
JKAIA 2025, NeurIPS preview session Seoul, Republic of Korea	Nov. 2025
SNU GSDS student research seminar Seoul, Republic of Korea	Apr. 2025
Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference	
SNU GSDS workshop poster session Jeju, Republic of Korea	Jul. 2025
NeurIPS 2024 poster session Vancouver, Canada	Dec. 2024
SNU GSDS student research seminar Seoul, Republic of Korea	Dec. 2024
Conference Reviewer: 2026 NeurIPS, UAI, ICLR, ICML, 2025 ICML	

REFERENCE

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